

Marketing Campaign Analysis Report

Customer Segmentation, Campaign Performance, Product Spending, and Channel Usage

Metric	Value
Total customers	56,000
Original core variables	28
Reporting dataset columns after features	44
Totals spend	35,858,736.00
Average income	57,252.19
Overall response rate	14.76%

Executive Summary

- The dataset contains 56,000 customer records with no missing values and no duplicate customer IDs after validation.
- Overall campaign response rate is 14.76%, meaning roughly one in seven customers responded to the latest campaign response flag.
- High spender customers represent 10.00% of customers and show a stronger response rate of 26.04% with average spend of 1,972.20.
- Very High income customers have the strongest income-band response rate at 25.36% and average spend of 1,141.32.
- Underserved customers total 9,677; they have low spend, high web visits, and no campaign response, making them a distinct reactivation group.

1. Problem Statement

A retail company has run multiple marketing campaigns and collected customer-level data covering demographics, product spending, purchase channel usage, web visits, campaign acceptance, and complaint behavior. Management needs a consolidated analytics solution to identify the best customers, understand what drives campaign response, and support future targeting decisions.

The project answers the following business questions:

1. Which customer segments have the highest response rate to marketing campaigns?
2. How do spending patterns differ by product category and customer profile?
3. Which channels are used most by high-value customers?
4. Which customer groups are underserved?
5. What profile should be targeted in future campaigns?

2. Dataset Overview

The cleaned working dataset contains 56,000 customer records. Each row represents one customer identified by ID. The raw business dataset uses 28 core variables, while the final reporting dataset contains 44 columns after derived features and segmentation flags.

Area	Variables / Description
Demographics	ID, Year_Birth, Education, Marital_Status, Income, Country
Family details	Kidhome, Teenhome, Children
Customer relationship	Dt_Customer, Recency, Customer_Tenure_Days
Product spending	Wines, Fruits, Meat, Fish, Sweets, Gold, Total_Spend
Channel usage	Deals, Web, Catalog, Store purchases, Monthly web visits
Campaign response	AcceptedCmp1 to AcceptedCmp5 and Response
Complaint	Complain

Dataset Metric	Value
Rows	56,000
Columns after feature engineering	44
Missing values	0
Duplicate customer IDs	0
Customer date range	2012-07-30 to 2014-06-29
Average income	57,252.19
Average total spend	640.33
Average total purchases	13.24
Average monthly web visits	5.17

3. Data Cleaning

The cleaning stage focused on making the dataset reliable before analysis, segmentation, SQL loading, and dashboard development.

- Loaded marketing campaign data and reviewed the data dictionary to understand field meanings.
- Checked missing values; total missing values found: 0.
- Checked duplicate customer IDs; duplicate IDs found: 0.
- Converted Dt_Customer into a date format for tenure calculation.
- Validated campaign columns as binary indicators with values representing acceptance or non-acceptance.

- Checked age and income ranges to avoid unrealistic customer profiles before segmentation.
- Standardized the analysis dataset by saving a cleaned CSV for SQL and dashboard use.

4. Feature Engineering

New fields were created to support customer behavior analysis and business segmentation.

Feature	Logic	Business Purpose
Age	2014 - Year_Birth	Used for age-band analysis and young-customer segmentation
Customer_Tenure_Days	Reference date - Dt_Customer	Measures customer relationship length
Children	Kidhome + Teenhome	Identifies family customers
Total_Spend	Sum of all product spending columns	Identifies high-value and high-spender customers
Total_Purchases	Sum of purchase channel columns	Measures customer purchase activity
Age_Band	Under 30, 30-45, 46-60, 60+	Supports demographic comparison
Income_Band	Low, Medium, High, Very High	Supports income-based response and spend analysis

5. EDA Findings

Exploratory data analysis was used to understand customer distribution, spend behavior, campaign response, and channel preference.

Product Category	Total Spend	Average Spend
Meat	15,024,465	268.29
Wines	13,830,963	246.98
Fish	2,861,473	51.10
Gold	1,981,595	35.39
Sweets	1,255,691	22.42
Fruits	904,549	16.15

Key EDA observations:

- Meat and wine are the two largest product spending categories in the dataset.
- Average spend increases strongly with income band, especially among Very High income customers.
- Older age bands show higher response rates than younger age bands.
- Website visits are common, but high visit volume does not always mean high campaign response.

Channel / Activity	Average per Customer
Monthly Web Visits	5.17
Store Purchases	4.71
Web Purchases	4.25
Deals Purchases	2.17
Catalog Purchases	2.11

6. Rule-Based Segmentation

Rule-based segmentation was applied using the business rules required for the project. These rules translate raw customer behavior into interpretable marketing groups.

Segment	Rule
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High Income	Income > 75,000
Young Customer	Age < 30
Campaign Responder	Response = 1
High Web Engagement	NumWebVisitsMonth > 5
Family Customer	Children > 0
High Spender	Total_Spend > 90th percentile
High Value Customer	High Spender OR High Income responder
Underserved Customer	Low spend + high web visits + no response

Segment	Customers	Response Rate	Avg Spend	Avg Income
Campaign Responder	8,265	100.00%	939.29	75,317.24
High Spender	5,599	26.04%	1,972.20	87,841.75
High Income	19,782	23.54%	1,038.03	94,997.93
High Web Engagement	29,970	12.07%	510.08	48,626.72
Young Customer	8,285	11.65%	485.58	51,779.08
Family Customer	38,222	11.62%	499.70	48,493.55

The strongest commercially useful segment is High Spender. It contains 5,599 customers, has an average spend of 1,972.20, and has a response rate of 26.04%. High Income customers are also valuable because they include 19,782 customers with 23.54% response rate and average spend of 1,038.03.

7. SQL Analysis

SQL scripts were created to make the analysis reproducible and to support the dashboard layer. The SQL layer uses a main customer table and analytical queries/views for KPIs and segment analysis.

SQL File	Purpose
create_tables.sql	Creates the marketing_customers table with demographic, spend, channel, campaign, and derived feature columns.
load_data.sql	Loads cleaned_marketing_data.csv into the marketing_customers table.
analytical_queries.sql	Contains KPI, campaign, segment, product, channel, underserved, ideal target, and ranking queries.

- Overall KPI query calculates total customers, average income, total spend, average spend, purchases, and response rate.
- Campaign acceptance query compares AcceptedCmp1 to AcceptedCmp5 and Response.
- Segment KPI query compares response rate, average income, spend, purchases, and web visits by segment.
- Product spend demographic query summarizes product spend by age and income band.
- Channel usage query compares high-spender and high-value customer behavior.
- Window function query ranks countries by response rate to demonstrate advanced SQL usage.

8. Dashboard Explanation

The dashboard was built using Streamlit and reads the cleaned dataset directly. It provides an interactive way to explore campaign performance, segment behavior, product spending, and channel usage.

Dashboard Section	Purpose
KPI Cards	Shows total customers, total spend, average income, average spend, and response rate.
Filters	Allows filtering by country, education, marital status, age band, and income band.
Campaign Performance	Compares acceptance rate across AcceptedCmp1 to AcceptedCmp5 and Response.

Customer Segments	Shows customer count and response rate by segment.
Product Spending	Compares total and average spend across product categories.
Channel Usage	Shows average usage of deals, web, catalog, store, and web visits.
Response Analysis	Shows response rate by income band, age band, and country.
Data Preview	Displays filtered customer records and allows CSV download.

Dashboard execution command:

```
cd dashboard
python -m pip install -r requirements.txt
python -m streamlit run app.py
```

9. Key Insights

Income Band	Customers	Response Rate	Avg Spend
Very High	12,518	25.36%	1,141.32
High	14,884	18.22%	790.93
Medium	13,863	11.77%	500.48
Low	14,735	5.07%	194.19

Age Band	Customers	Response Rate	Avg Spend
60+	4,916	18.96%	860.75
46-60	16,427	16.69%	743.71
30-45	26,372	13.75%	583.48
Under 30	8,285	11.65%	485.58

Top Country	Customers	Response Rate	Avg Spend
Australia	5,677	20.19%	827.38
India	4,814	18.76%	723.85
Germany	4,432	16.81%	707.24
USA	4,331	15.70%	663.88
Mexico	875	15.20%	634.14

- Very High income customers have the strongest response rate by income band at 25.36%.
- Customers aged 60+ have the highest age-band response rate at 18.96%.
- Australia has the strongest country response rate among listed countries at 20.19%.
- High spenders show substantially higher commercial value than the full customer average.
- Underserved customers total 9,677; their average monthly web visits are 7.27, showing digital activity without conversion.

10. Recommendations

6. Prioritize high-spender customers for premium bundles and loyalty offers because their response rate and average spend are both above the overall average.
7. Target Very High and High income customers with higher-value campaign offers, since response and spend are stronger in these bands.
8. Use campaign responders as a remarketing audience because previous response behavior is the clearest signal of future campaign receptiveness.
9. Develop web-based campaigns for high web engagement customers, but separate high visit volume from actual conversion behavior.

10. Create reactivation offers for underserved customers, using low-risk discounts or product trials because they visit frequently but have low spend and no response.
11. Use country and age-band filters before campaign launch because response rates differ materially by country and age group.

11. Conclusion

This project created a complete marketing analytics solution using Python, SQL, and Streamlit. Python was used for cleaning, EDA, feature engineering, and segmentation. SQL was used for table creation, loading logic, analytical queries, dashboard-supporting views, and ranking analysis. Streamlit was used to provide an interactive dashboard for stakeholder exploration.

The analysis shows that campaign strategy should not treat all customers equally. High spenders, high-income customers, prior responders, and specific age-country segments offer stronger campaign potential. Underserved customers require separate reactivation treatment because they show website activity without campaign conversion. These findings support a more targeted and commercially practical campaign strategy.

Appendix: Final Deliverables

Deliverable	Location
Cleaned dataset	data/cleaned_marketing_data.csv
Python notebook	notebooks/marketing_campaign_analysis.ipynb
SQL scripts	sql/create_tables.sql, sql/load_data.sql, sql/analytical_queries.sql
Dashboard	dashboard/app.py and dashboard/requirements.txt
Report	reports/marketing_campaign_analysis_report.pdf
README	README.md